

WEEK 6 – HEALTHCARE ANALYTICS DASHBOARD

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Introduction:

Week 6 focuses on creating DAX measures and developing an interactive healthcare analytics dashboard using the healthcare dataset prepared in Week 5. The existing data model was used as the base, and a separate Measures Table Healthcare was created to store the DAX measures used in the dashboard.

The dashboard provides information about total patients, billing amount, average billing amount, average length of stay, medical conditions, admission types, insurance providers, and patient trends over time.

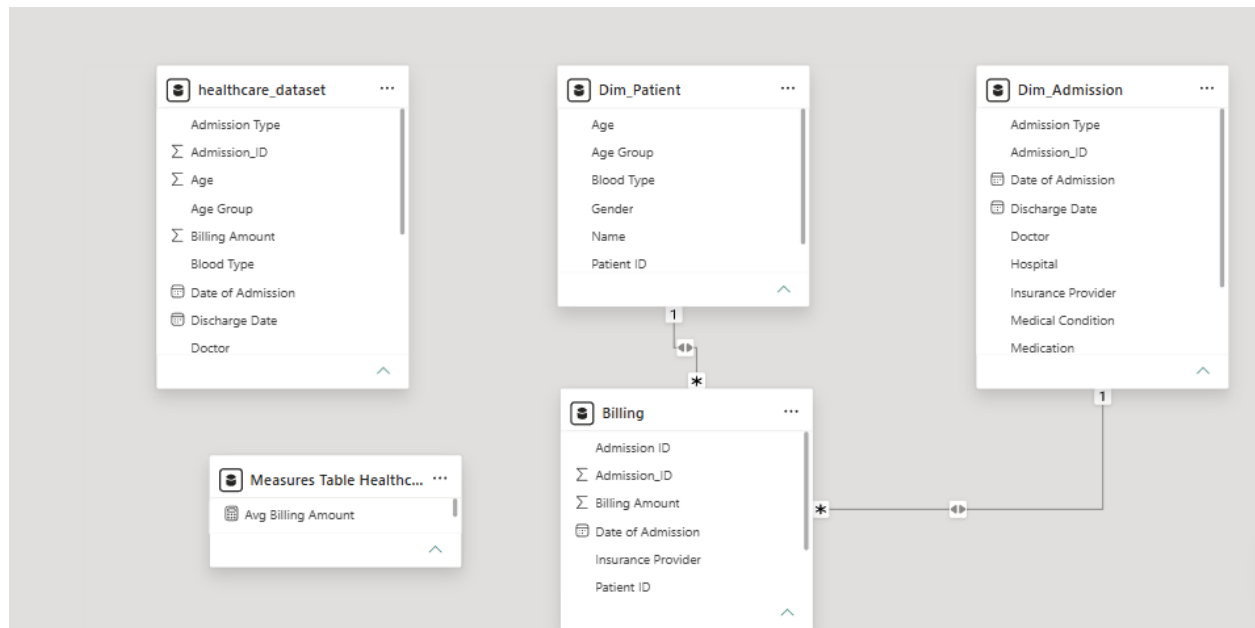
Data Model:

The Week 5 healthcare data model was used as the foundation for Week 6. The model contains the following tables:

- Dim_Patient
- Dim_Admission
- Billing
- Healthcare Dataset
- Measures Table Healthcare

A separate Measures Table Healthcare was created to organize and store the DAX measures used in the dashboard.

The data model allows the different patient, admission, and billing information to be analyzed together through the existing relationships.

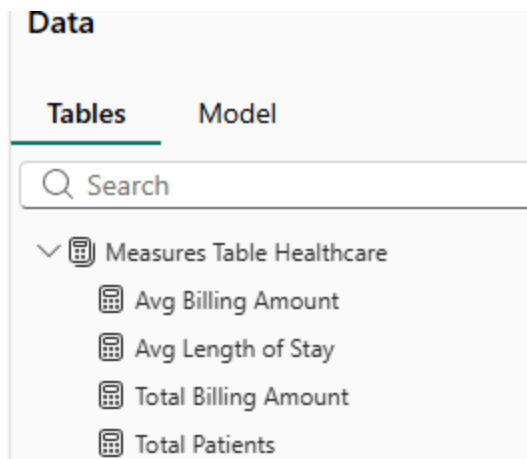


Measures Table:

A separate Measures Table Healthcare was created to store the DAX measures. This helps keep the calculated measures organized and makes them easier to use throughout the dashboard.

The following measures were created:

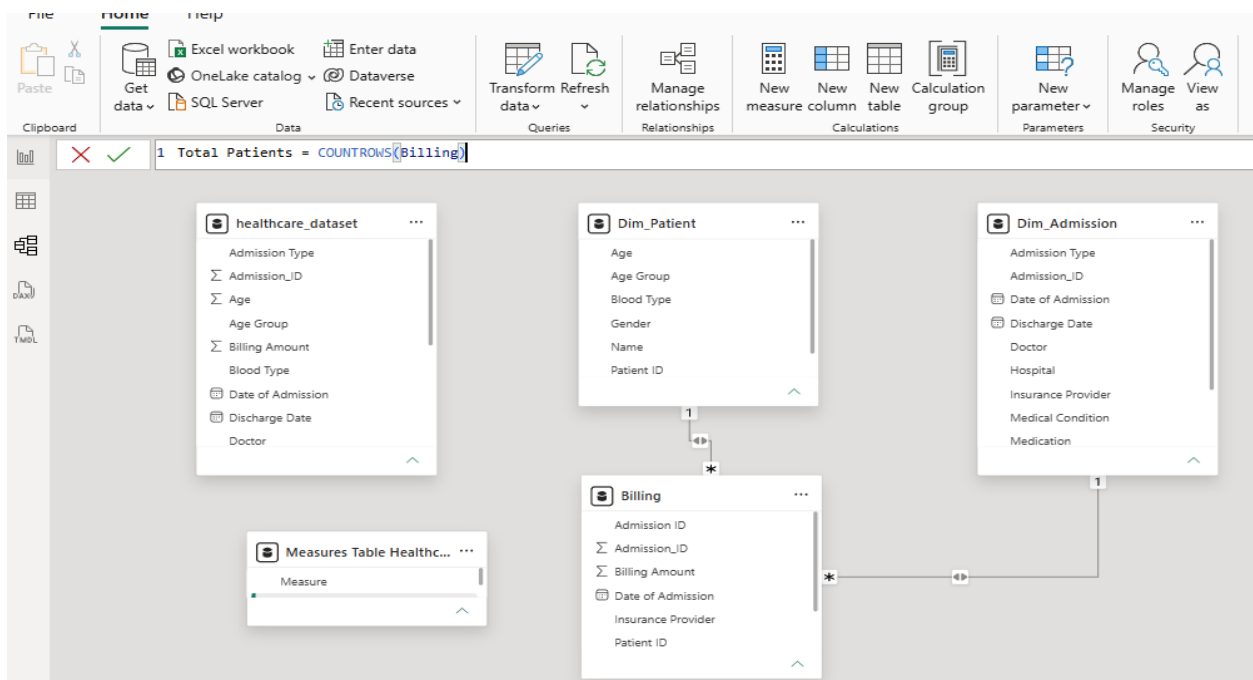
1. Total Patients
2. Total Billing Amount
3. Avg Billing Amount
4. Avg Length of Stay



DAX Measures:

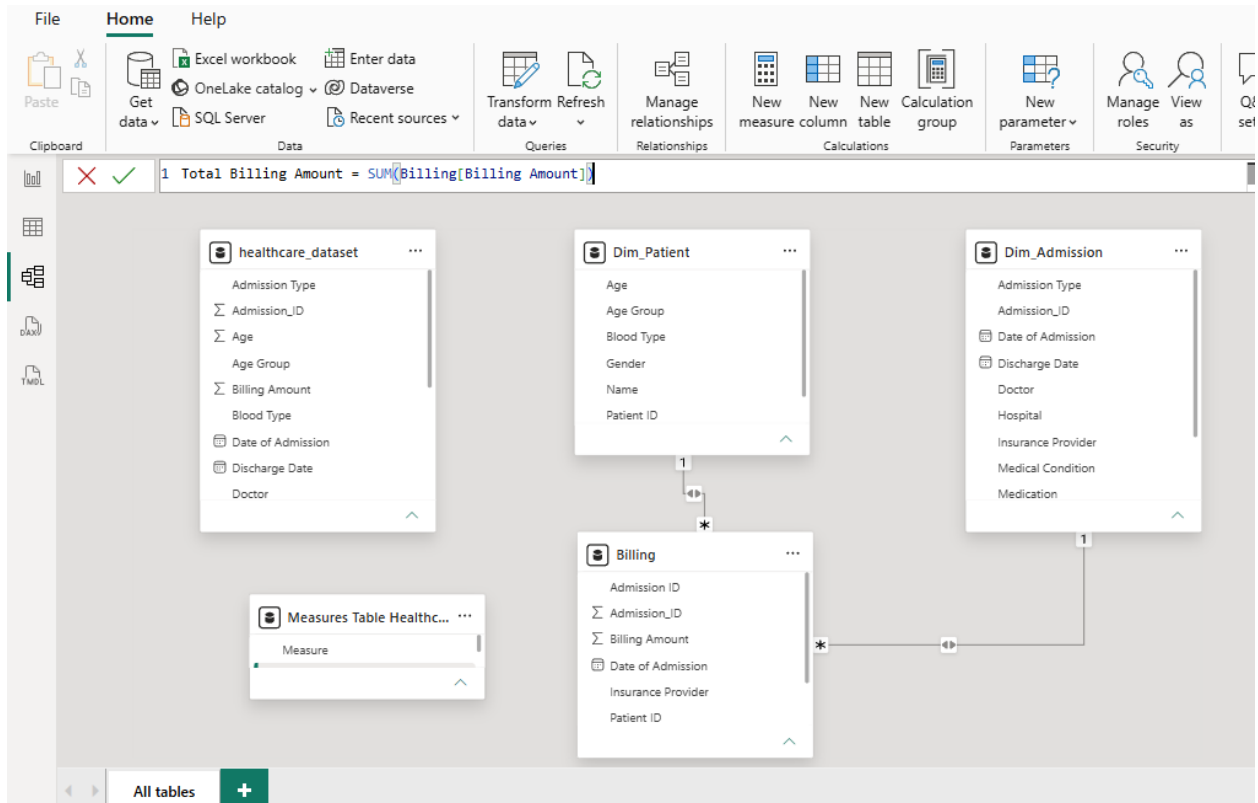
TOTAL PATIENTS

The Total Patients measure was created using the COUNTROWS function on the Billing table. It calculates the total number of records available in the billing data.



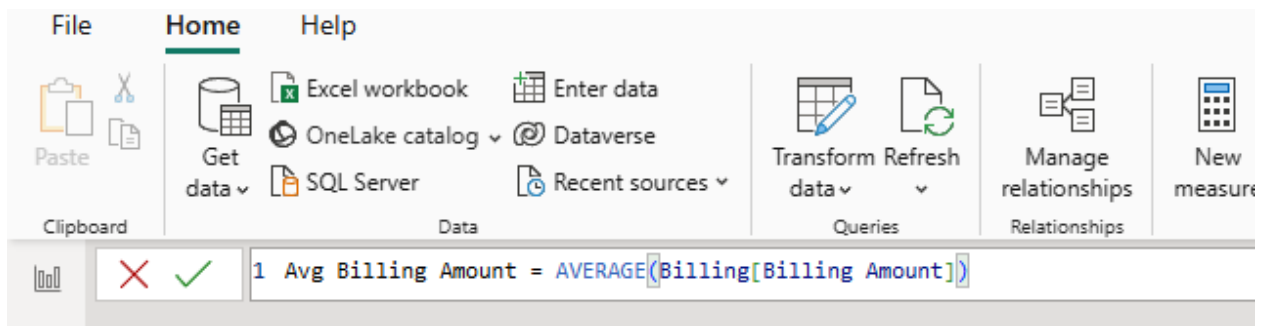
Total Billing Amount:

The Total Billing Amount measure was created using the SUM function on the Billing Amount column. It calculates the total billing amount.



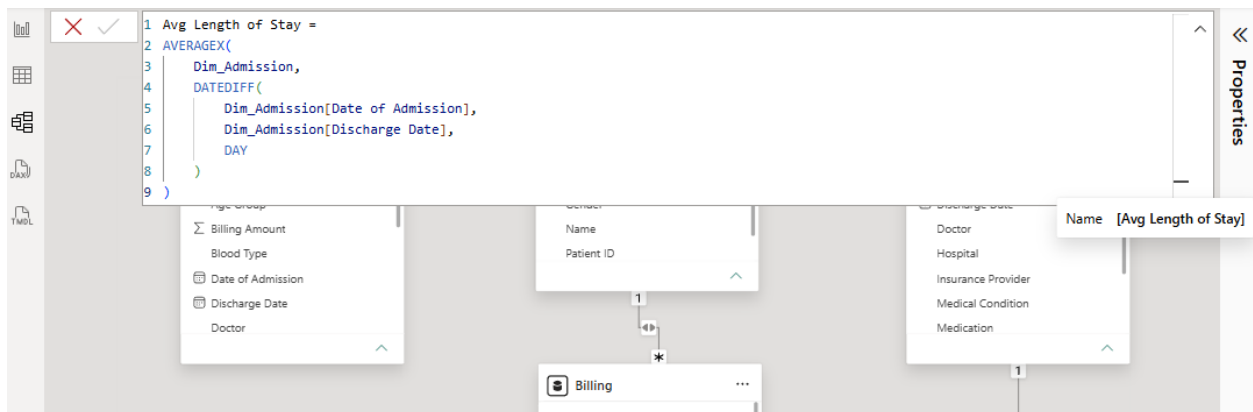
Average Billing Amount:

The Avg Billing Amount measure was created using the AVERAGE function on the Billing Amount column. It calculates the average billing amount across the available billing records.



Average Length of Stay:

The Avg Length of Stay measure was created using AVERAGEX and DATEDIFF. It calculates the number of days between Date of Admission and Discharge Date and then finds the average length of stay.



Dashboard Cards:

Four card visuals were created to display the main healthcare metrics:

- Total Patients
- Total Billing Amount
- Average Billing Amount
- Average Length of Stay

These cards provide a quick overview of the main values in the healthcare dataset.

Matrix:

A matrix visual was created to analyze patients based on Age Group, Gender, and Medical Condition. This allows patient counts to be compared across different age groups, genders, and medical conditions.

Donut Chart:

A donut chart was created to show the distribution of total patients across different age groups. The chart provides a visual comparison of patient counts among the age groups.

Medical Condition by Admission Type:

A stacked bar chart was created to analyze the number of patients for each medical condition based on admission type. The admission types included Elective, Emergency, and Urgent.

This visualization makes it easier to compare patient counts across different medical conditions and admission types.

Slicers and Dashboard Interactivity:

Three slicers were added to make the dashboard interactive:

- Insurance Provider
- Admission Type
- Date of Admission

The slicers allow users to select specific values and filter the dashboard visuals accordingly. This makes it easier to analyze healthcare information based on different criteria.

Final Healthcare Dashboard:

The final dashboard combines the DAX measures with different Power BI visuals to provide an interactive view of the healthcare data.

The dashboard includes KPI cards, a matrix, donut chart, stacked bar chart, line and stacked column chart, clustered column chart, and interactive slicers.

The dashboard can be used to analyze patient counts, billing amounts, average billing, length of stay, medical conditions, admission types, insurance providers, and trends over time.



Conclusion:

Week 6 focused on creating DAX measures and developing an interactive healthcare analytics dashboard in Power BI. The dashboard combines

patient, admission, billing, medical condition, and insurance information into different visualizations.

The created measures provide important summary information such as total patients, total billing amount, average billing amount, and average length of stay. The different charts and slicers make it easier to explore the healthcare data and analyze it from different perspectives.

Overall, the Week 6 dashboard provides an organized and interactive way to understand the healthcare dataset.